# ShopSteward

## Project proposal

### A personal operations agent for small merchants

Project proposal | 8 September 2026

### Project overview

A promotion is approaching. The owner needs enough stock to meet demand, but every purchase leaves less cash for the rest of the business. Once an order is placed, the work continues: stock must arrive, sales change what is needed, and a late delivery can undo a sensible plan.

ShopSteward is a proposed operations agent that follows this work through. The owner can ask, "Keep this promotion stocked while preserving my cash reserve." The assistant compares purchase options and asks for confirmation. It then follows the order, checks changing conditions after the conversation ends and returns with a new decision when the owner needs to act.

The project brings language-model tool use together with business rules and informed search. Explicit corrections can also become reusable procedures for later tasks. A controlled business simulation will make the consequences visible: what was bought, whether the reserve held, and whether the assistant followed through. On returning, the owner should see what happened and what still needs a decision.

### Group information

Canvas group number _____50_____


| Member | Full name | NUS student ID |
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# 1 The merchant and the opportunity

A small merchant often serves customers and manages purchasing in the same working day. A stock decision competes for attention with the very sales it is meant to support. The owner may know the business well, yet still have to assemble the latest cash balance, shelf count and supplier information before deciding what to buy.

The decision also has a memory. Forty units ordered yesterday may cover the shortage visible today. Sales recorded today may not become spendable cash until later. A demand revision can make yesterday's purchase insufficient. Each new recommendation therefore depends on what has already happened and what is still outstanding.

## Evidence and existing products

Forthcast's 2026 interviews with 13 Shopify operators recorded manual spreadsheet planning in 11 interviews and inventory cash or cash-visibility concerns in six. This vendor-run qualitative study provides a signal from a small recruited panel; its counts cannot be projected to all merchants [4].

| Existing approach | Relevant capability | Implication for ShopSteward |
|---|---|---|
| Shopify Sidekick [5] | Sales-based replenishment advice, purchase orders and receiving workflows. | Conversational purchasing is an established capability. |
| Prediko [6] | Demand and supply plans connected to incoming stock, budgets and purchase orders. | A useful proposal must address more than the next order quantity. |
| Business record systems [3] | Point-of-sale and integrated business records. | Reliable assistance needs current, traceable inputs. |

ShopSteward will examine the value of keeping a delegated goal active across conversations and business events, while retaining an explicit correction for later work. The comparison is a test of these behaviours in a defined setting; it does not assume that existing products lack them.

## A focused first user

The target is an owner who controls purchasing and sells non-perishable goods with identifiable stock, costs and quantities. This can include an online seller or a neighbourhood retailer. Enterprise Singapore's support for point-of-sale and business systems makes offline retail a relevant product context [3], although demand for this particular assistant remains untested.

The first demonstration centres on one shop preparing for a promotion. Its intended benefit is less repeated explanation and manual checking: the owner sees what changed, why a decision is needed and what will happen after confirmation. Real-user adoption will require subsequent field evaluation.

# 2 A delegated task in action

Consider a constructed shop scenario. The owner has 20 units in stock, expects demand for 60 and holds 1,000 currency units (CU). A unit costs 10 CU to buy and sells for 20 CU. The owner asks ShopSteward to prepare for the promotion while keeping at least 300 CU available.

The assistant compares quantities of 0, 20, 40 and 80. Buying 40 covers the shortage and leaves 600 CU; buying 80 would leave only 200 CU and is excluded. The owner sees the quantity, total cost and cash effect, then confirms that specific purchase in the interface.

## What happens after confirmation

The owner can close the conversation. ShopSteward retains the task and checks the order while it is in transit. Those 40 incoming units count towards the need, so another check should recommend waiting. After delivery and a sale of 10 units, a demand update raises the remaining requirement from 50 to 70. The assistant then explains the new gap and proposes a further 20 units for fresh confirmation.

| Event | Cash | Stock | In transit | Unsettled sales |
|---|---|---|---|---|
| Initial state | 1,000 | 20 | 0 | 0 |
| Confirm and order 40 | 600 | 20 | 40 | 0 |
| Receive 40 | 600 | 60 | 0 | 0 |
| Sell 10 | 600 | 50 | 0 | 200 |
| Raise remaining demand to 70 | 600 | 50 | 0 | 200 |
| Confirm and order another 20 | 400 | 50 | 20 | 200 |
| Receive the additional 20 | 400 | 70 | 0 | 200 |

Table 1  Expected states for a synthetic acceptance case. Cash and unsettled sales are in CU; stock and in-transit quantities are units.

Purchasing reduces cash when the simulated order is accepted; receiving moves units into stock. Sales proceeds remain unsettled throughout this case. The final row is a stocking checkpoint, with subsequent sales and settlement still outstanding. A brief will make these distinctions visible alongside the task history and next check.

## Delivery scope

The purchasing demonstration uses one merchant, one item and one supplier. The assistant also supports business queries, background follow-up, task recovery and a quotation procedure reused on a new file. Purchasing and payments occur only in simulation. Multiple merchants, perishables, pricing, marketing and live store connections are outside the proposed delivery.

# 3 System design and reasoning

The language model interprets the owner's goal, calls bounded business tools and explains their results. Business services calculate options and enforce the rules. The interface sends the owner's confirmation directly to those services. The assistant can request a revised proposal; it has no tool that independently approves or places a purchase.

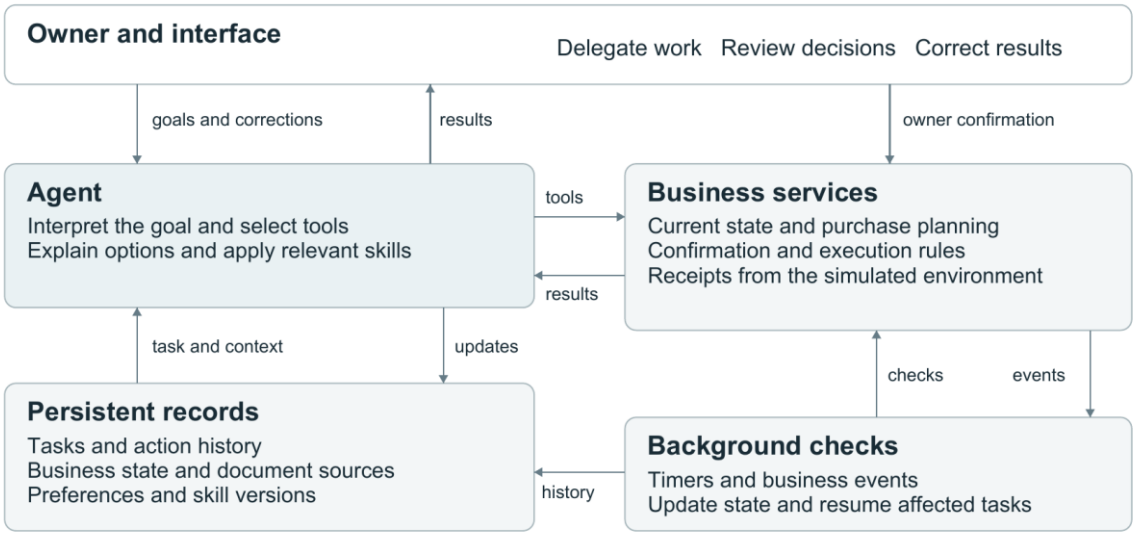



Figure 1 Logical responsibilities. The confirmation path reaches business services directly. Persistent records support both background checks and the assistant's next interaction.

| Course technique group | Role in the proposed system |
|---|---|
| Decision automation | Business rules enforce the cash reserve, current confirmation and one purchase per approved action. |
| Cognitive techniques | Language-model tool use connects natural-language goals with task context, explanations and saved procedures. |
| Business resource optimisation | Informed search compares purchase sequences. Beam search is the proposed method, evaluated against a daily rule and enumeration. |

These three groups connect through the same task and business state, as required by the course brief [1]. The planner returns numerical options; the model helps the owner interpret them; execution rules remain in force regardless of which option is proposed.

A confirmation is bound to the item, quantity, amount, proposal and relevant state. A material change requires reassessment and a new confirmation. Retries reuse the original action identity; an uncertain response triggers a receipt check. Timers and business events resume the persistent task after records are updated. Closing a conversation preserves the task; stopping follow-up preserves its history and still resolves an action already in progress.

# 4 Personalisation and information

An owner should be able to improve how the assistant works through an ordinary correction. "Show cash and stock first" is a presentation preference. "Compare quotations by unit price, and keep the pack size and source" changes the procedure used to process a file. ShopSteward will retain these changes with their intended use.

## A correction that survives the next task

In a constructed quotation example, the first file lists 120 CU for a box of 12 units and a minimum order of two boxes. The owner asks for comparison by unit price while retaining pack size, minimum order and source. The assistant should show 10 CU per unit and propose a saved procedure for making that conversion.

A later conversation supplies a different item: 180 CU per box of 15, with a minimum order of three boxes. Applying the procedure must produce 12 CU per unit from the new file. The reusable element is the processing method; the new price and quantities come from the new input.

A skill is a saved procedure defining when it applies, which tools to use and what the output must contain. An explicit correction creates a version linked to its source task. Before activation, the system checks the revision against the corrected example and a missing-field case. Subsequent runs record the selected version and their tool results. Such revisions cannot change purchasing permissions or the cash reserve.

## Information with distinct roles

The business ledger supplies current cash, stock and outstanding orders. Reference documents retain evidence such as a supplier's pack size or terms, with the source and version available for inspection. Preferences describe the owner's requested presentation; skills describe processing steps. A document or remembered statement cannot silently replace a ledger value.

The core quotation workflow uses comma-separated values (CSV) files containing currency, price basis, pack size, minimum-order units and source row. Missing pack size blocks conversion of a box price; an explicit unit price can still be reported. Processing a quotation produces a traceable result without updating stock or submitting an order. Broader document retrieval can be evaluated separately from this workflow.

## Data preparation

A controlled simulator supplies versioned initial states and dated purchase, delivery, sales and demand-update events. Demand is a scenario input; this proposal does not depend on training a forecasting model. Quotation files and reference answers will be constructed separately, checked for units and missing fields, and split into development and held-out sets. Future events become visible only when released. Reference answers remain separate from the assistant's input and are used only by the evaluator.

# 5 Purchasing across several periods

Buying the right total quantity still leaves a timing decision. An early order protects availability but holds stock and commits cash sooner. A later order can preserve flexibility, provided it arrives before the demand. The proposed planner will compare these consequences over several decision periods.

## A simple timing example

Consider a separate three-day illustration with 20 units in stock, 1,000 CU cash, a 300 CU reserve and a unit purchase cost of 10 CU. Demand is 10, 10 and 40 units. Orders arrive before sales the following day; sales proceeds remain unsettled. Both plans below serve all 60 units.

| Purchase plan | Cash after day 1 | Closing stock on days 1 2 3 | Stock carried |
|---|---|---|---|
| Order 40 on day 1 | 600 CU | 10, 40, 0 | 50 unit-days |
| Order 40 on day 2 | 1,000 CU | 10, 0, 0 | 10 unit-days |

Table 2  Calculated illustration, not algorithm output. Stock carried is the sum of closing daily inventory; ten units held for one day are ten unit-days.

The planner first minimises projected unmet demand, then stock carried, with smaller total purchase quantity breaking remaining ties. Every candidate must preserve the cash reserve and respect delivery timing. Existing orders remain committed; only the next purchase is recommended for confirmation. Later choices are reconsidered as new information arrives.

## Why compare search with simpler methods

| Method | Purpose in the comparison | Limitation |
|---|---|---|
| Daily replenishment rule | A cheap, explainable next-order baseline. | Does not retain alternative purchase sequences. |
| Exhaustive enumeration | An exact reference under shared planning assumptions. | Sequence count grows with the horizon and permitted quantities. |
| Beam search | A bounded search over alternative partial plans. | May discard the best plan; ranking and beam width matter. |

Beam search is an informed search strategy covered in Machine Reasoning [8]. At each period it expands permitted quantities, rejects constraint violations and completes each partial plan with the daily rule to estimate the consequences. It retains a limited number of the best-ranked partial plans. Beam width is selected on development cases.

Four quantities over seven days produce 16,384 sequences before pruning, small enough to support an exact reference comparison. This experiment will test the quality and runtime tradeoff of bounding the search, rather than presume that search beats a simple rule. Research on inventory control motivates examining language models and operations research together [7]; their effects will be measured separately here.

# 6 Evaluation questions and measures

The central test is whether the owner can leave a delegated task and return to work that has been carried out correctly. A test episode follows one task to a specified endpoint, including any required waiting, confirmation and resolution of pending actions. The fixed states in Table 1 and the quotation examples establish correctness before comparative evaluation.

| Measure | Definition and interpretation |
|---|---|
| Task success rate | Successful episodes / attempted episodes. All expected checkpoints, a retrievable result and no execution violation are required. |
| Unmet demand | Sum of daily max(0, realised demand minus stock available after receipts and before sales). Unserved units are lost sales. |
| Stock carried | Sum of closing daily inventory in unit-days, assessed alongside unmet demand. |
| Execution violations | Unauthorised, repeated or reserve-breaching purchases. Required count in the acceptance suite is zero. |
| Quotation accuracy | Correct required fields / evaluated fields. Decimal unit prices are checked exactly; a genuinely missing field marked unavailable is correct. |
| Planning gap and cost | Excess projected unmet demand versus enumeration; on a tie, excess projected unit-days. Record runtime, model calls, tokens and cost per episode. |

### Comparisons that isolate the contribution

For planning, a scripted controller compares the daily rule, Beam search and enumeration using identical visible demand assumptions and daily replanning times. The daily rule chooses the feasible quantity with the least forecast shortage before the next daily order could arrive, counting stock already in transit; ties use the smaller quantity.

For response timing, the planner is held fixed at Beam search. Daily replanning is compared with daily replanning plus a check when a demand or delivery update arrives. A scripted controller isolates this timing effect from language-model behaviour. Both conditions enforce the same confirmation and execution rules.

For the assistant, completion and recovery checks compare the model-driven workflow with a scripted workflow using the same business tools, storage and scheduler. The checks include a real timed wake-up without a new user message, a business event, a new conversation and a process restart. Prescribed waiting and the absence of a duplicate purchase are part of success.

For procedure reuse, the same model and tools process new quotations with and without the revised skill, while display preferences remain fixed. Each file starts an independent task with identical base context. The control excludes the correction conversation and any memory containing that revision. The skill is frozen before held-out testing, with no learning between test files.

# 7 Study protocol and delivery

The proposed evaluation budget is 30 independent seven-day business scenarios: ten under stable conditions, ten with a demand revision and ten with a delivery delay. Initial stock, usable cash, demand and event times will vary. A further 30 held-out quotation files will cover box prices, unit prices and missing fields. Five development scenarios and separate quotation examples are excluded from final results.

These are planning targets. A development pilot will establish feasible runtime and model cost; the final sample sizes, beam width and configurations will then be frozen before held-out evaluation. The evaluation record will preserve that specification alongside software, model, prompt, data and skill versions.

## A shared environment

Each simulated day begins with receipts and scheduled planning. Demand or delivery updates are then released, followed by the extra check in the event-driven condition and then sales. Future information is withheld until release. A delivery disruption applies through a shared supplier time window to affected outstanding and newly placed orders. Its notification precedes each affected receipt, so different policies face the same supply conditions.

All policies share the initial conditions, external shocks, permitted quantities and cash rules. The batch evaluator approves only current proposals that pass cash, version and uniqueness checks; the interactive demonstration requires the owner's explicit confirmation. Model-based conditions are planned for three repeats, averaged within each case. Paired differences will use 95% bootstrap intervals over independent cases, not individual repeats. Timeouts count as failures, and costs include retries.

## What the evidence can establish

A successful demonstration would show a task surviving the end of a conversation, reacting appropriately to a change and preserving the owner's control over purchases. A successful procedure test would show a correction changing a later result on genuinely different input. Comparative experiments will determine whether these behaviours improve outcomes enough to justify their cost; an inconclusive or unfavourable result remains informative.

The simulation makes purchases and failures observable without a live merchant account. It simplifies operations to non-perishable stock with known unit prices and scenario-supplied demand. Sales proceeds remain unsettled over the evaluation horizon. Closing stock, pending deliveries and unsettled proceeds stay visible at the endpoint; closing cash is not treated as profit. Real merchant income, adoption and willingness to delegate require later field evidence.

## Proposed delivery

Development will first establish the purchasing and follow-up path, then verify correction reuse on new quotation files, before running the controlled comparisons. The final package will contain a runnable system, a report with methods and findings, reproducible configurations and a user guide, together with the course's two five-minute videos covering business use and system design [1].

The project will demonstrate what it takes for a merchant to entrust an assistant with continuing work: a goal that remains active, decisions grounded in current records, and a clear account of what happened when the owner was elsewhere.

# References

[1] NUS ISS. IRS practice module and exam briefing, v2.17, 2026. Technique groups and final delivery requirements, PDF pp. 6 and 9-14. Course briefing

[2] NUS ISS. IRS practice module project proposal and final presentation guidelines, v016, 2026. Proposal content and optional sections, PDF pp. 2-14. Proposal guidelines

[3] Enterprise Singapore. Business Productivity Solutions. Heartland enterprise support and examples of business systems. Accessed 8 September 2026. Enterprise Singapore

[4] Forthcast. State of Shopify Inventory Operations 2026. Published 18 July 2026; interviews conducted February-March 2026. Study and methodology

[5] Mavadia, A. Managing inventory in Shopify just got easier. Shopify, 17 June 2026. Shopify inventory workflow

[6] Sujith, A. Planning with the Demand to Supply to Buy flow. Prediko Help Center, 25 June 2026. Prediko planning workflow

[7] Baek, J., Fu, Y., Ma, W. and Peng, T. AI Agents for Inventory Control: Human-LLM-OR Complementarity. arXiv:2602.12631v2, 4 May 2026. Research paper

[8] NUS ISS. Machine Reasoning Day 1, JR v1, 2026. Informed search strategies and evaluation functions, PDF pp. 46-47 (slide labels 51-52). Course handout.

## AI assistance

AI tools assisted source review, drafting, language editing and production of the diagram and document layout. The submitting team is responsible for checking citations, validating the system and explaining the work. Presentation speech will comply with the course restriction on AI-generated speech [1].